

Chapter 104

Generator Nodes

This chapter details the Generator nodes available in Fusion.

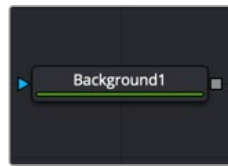
The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Background [Bg]



The Background node

Background Node Introduction

The Background node can be used to produce anything from a solid color background to complex looped gradients.

Inputs

There is one input on the Background node for an effect mask input.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the background color to only those pixels within the mask.

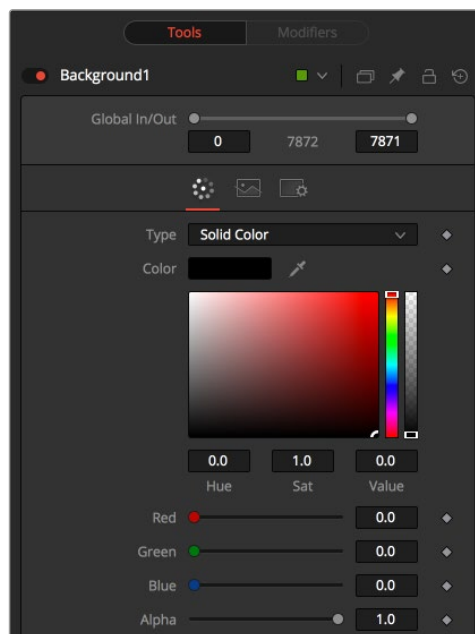
Basic Node Setup

The Background node generates a color and passes the color off to another node like the Paint node below.



A Background node generating a background for the Paint tool

Inspector



The Background node controls

Color Tab

The Color tab is used to control the style and color(s) of the generated background.

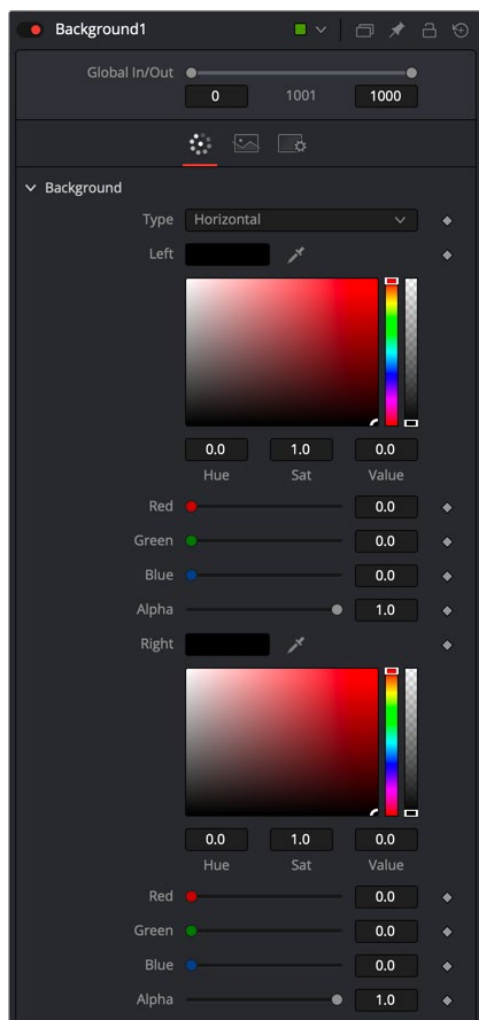
Type

This control is used to select the style of background generated by the node. Four selections are available:

- **Solid Color:** This default creates a single-color image.
- **Horizontal:** This creates a two-color horizontal gradation.
- **Vertical:** This creates a two-color vertical gradation.
- **Four Corner:** This creates a four-color corner gradation.

Horizontal/Vertical/Four Point

When the Type menu is set to Horizontal, Vertical, or Four Corner, two- or four-color swatches are displayed where the left/right, top/bottom, or four corners of the gradient colors can be set



Horizontal gradient controls



Gradient controls

Gradient

When the Type menu is set to Gradient, additional controls are displayed where the gradient colors' direction can be customized.

Gradient Type

This menu selects the form used to draw the gradient. There are six choices:

- **Linear:** Draws the gradient along a straight line from the starting color stop to the ending color stop.
- **Reflect:** Draws the gradient by mirroring the linear gradient on either side of the starting point.
- **Square:** Draws the gradient by using a square pattern when the starting point is at the center of the image.
- **Cross:** Draws the gradient using a cross pattern when the starting point is at the center of the image.
- **Radial:** Draws the gradient in a circular pattern when the starting point is at the center of the image.
- **Angle:** Draws the gradient in a counterclockwise sweep when the starting point is at the center of the image.

Viewer Start and End Position

The Start and End positions in the viewer are represented by two red control points connected by a green line. They determine where the gradient begins and ends.

Gradient Colors

This gradient color bar is used to select the colors for the gradient. The default two color stops set the start and end colors. You can change the colors used in the gradient by selecting the color stop, and then using the Eyedropper or color swatch to set a new color.

You can add, move, copy, and delete color from the gradient using the gradient bar.

To modify one of the colors, select the triangle below the color on the bar.

To add a color stop to the gradient bar:

- 1 Click anywhere along the bottom of the gradient bar.
- 2 Use the Eyedropper or color swatch to select the color for the color stop.

To move a color stop on the gradient bar:

Drag a color stop left or right along the bar.

To copy a color stop:

Hold Command (macOS) or Ctrl (Windows) while you drag a color stop.

To delete a color stop:

Drag the color stop up past the gradient bar.

Interpolation Space

This menu determines what color space is used to calculate the colors between color stops.

Offset

The Offset control is used to offset the position of the gradient relative to the start and end markers. This control is most useful when used in conjunction with the repeat and ping-pong modes described below.

Repeat

This menu includes three options used to set the behavior of the gradient when the Offset control scrolls the gradient past its start and end positions. Selecting Once keeps the color continuous

for offset. Selecting Repeat loops around to the start color when the offset goes beyond the end color. Selecting Ping-pong repeats the color pattern in reverse.

Sub-Pixel

The Sub-Pixel menu controls the sub-pixel precision used when the edges of the gradient become visible in repeat mode, or when the gradient is animated. Higher settings will take significantly longer to render but are more precise.

Gradient Contextual Menu

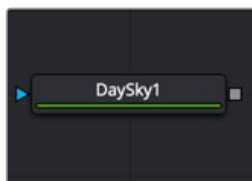
Gradients have their own contextual menu that you can bring up by right-clicking on the gradient bar. The Gradient contextual menu includes options for animating, publishing, and connecting one gradient to another. There is also a gradient-specific modifier that builds a custom gradient by sampling colors from the output of a node.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are duplicated in many Generator nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Day Sky [DS]



The Day Sky node

Day Sky Node Introduction

The Day Sky node aims to produce a simulation of the daylight produced at a specific time and location on the earth, and generates a high dynamic range image that represents a map of that light. It is not a sky generator, although it could be combined with a cloud generator or Noise node to produce one.

NOTE: This generator is a practical implementation of the research paper, *A Practical Analytical Model for Daylight*, by Preetham, Shirley, and Smits. A copy of the original paper can be found at the website for the Visual Simulation Group at the University of Utah [<https://www.cs.utah.edu/~shirley/papers/sunsky/sunsky.pdf>].

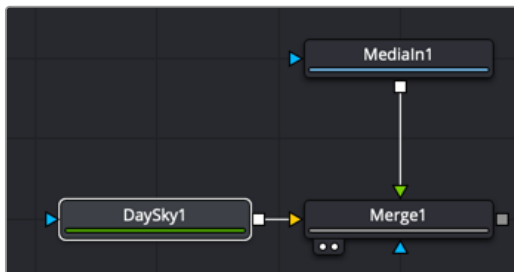
Inputs

There is a single input on the Day Sky node for an effect mask to limit the area where the day sky simulation occurs is applied.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Day Sky to only those pixels within the mask.

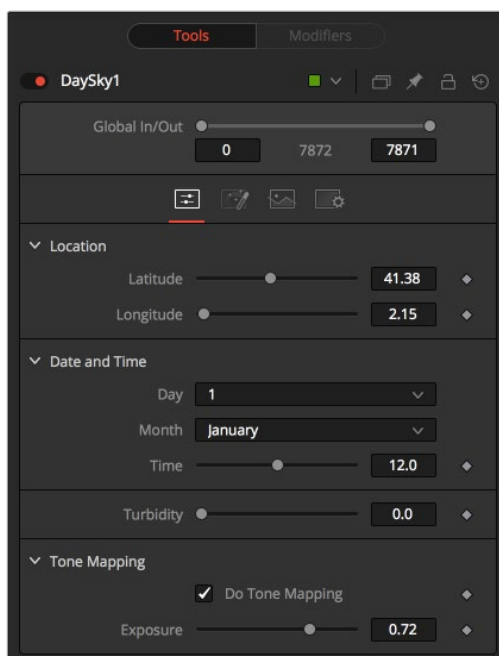
Basic Node Setup

The Day Sky node is a generator, so it typically starts the branch of a node tree and connects to some other node, like a Merge node.



A Day Sky connected as the background to a Merge node

Inspector



Day Sky controls

Controls Tab

The Controls tab is used to set the location and time of the daylight simulation. This will determine the overall look that is generated.

Location

The Latitude and Longitude sliders are used to specify the location used to create the Day Sky simulation.

Date and Time

The Day, Month, and Time controls are used to select the specific time for the Day Sky simulation.

Turbidity

Turbidity causes light to be scattered and absorbed instead of transmitted in straight lines through the simulation. Increasing the turbidity will give the sky simulation a murky feeling, as if smoke or atmospheric haze were present.

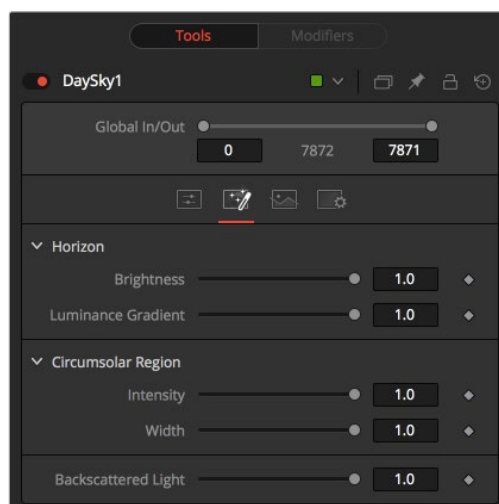
Do Tone Mapping

Since the simulation is calculated in 32-bit floating-point color space, it generates color values well above 1.0 and well below 0.0. Tone mapping is a process that takes the full dynamic range of the resulting simulation and compresses the data into the desired exposure range while attempting to preserve as much detail from the highlights and shadows as possible. Deselect this checkbox to disable any tone mapping applied to the simulation.

Generally, this option should be deselected only if the resulting image will later be color corrected as part of a floating-point color pipeline.

Exposure

Use this control to select the exposure used for tone mapping.



Day sky Advance tab controls

Advanced Tab

The Advanced tab provides more specific controls over the brightness and width of the different ranges in the generated sky.

Horizon Brightness

Use this control to adjust the brightness of the horizon relative to the sky.

Luminance Gradient

Use this control to adjust the width of the gradient separating the horizon from the sky.

Circumsolar Region Intensity

Use this control to adjust the intensity or brightness of the sky nearest to the sun.

Circumsolar Region Width

Use this control to adjust the width or size of the area in the sky affected by the sun.

Backscattered Light

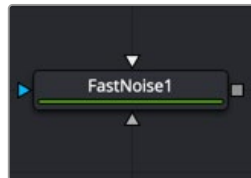
Use this control to increase or decrease the backscatter light in the simulation.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are duplicated in many Generator nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Fast Noise [FN]



The Fast Noise node

Fast Noise Node Introduction

The Fast Noise node is a very fast and flexible Perlin Noise generator. It can be useful for a wide range of effects, from clouds, to swirling fog, waves, water caustics, stylized fire, and smoke, and other organic textures. It is also invaluable as a noise source for other effects, such as heat shimmers, particle systems, and dirtiness maps.

Inputs

The two map inputs on the Fast Noise node allow you to use masks to control the value of the noise detail and brightness controls for each pixel. These two optional inputs can allow some interesting and creative effects. There is also a standard effect mask input for limiting the Fast Noise size.

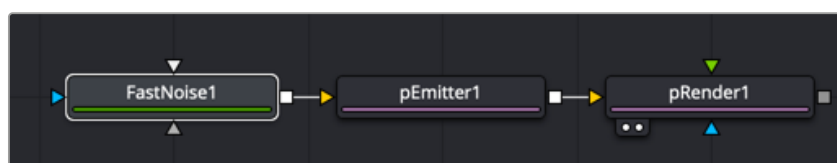
Noise Detail Map: A soft-edged mask connected to the gray Noise Detail Map input will give a flat noise map (zero detail) where the mask is black, and full detail where it is white, with intermediate values smoothly reducing in detail. It is applied before any gradient color mapping. This can be very helpful for applying maximum noise detail in a specific area, while smoothly falling off elsewhere.

Noise Brightness Map: A mask connected to this white input can be used to control the noise map completely, such as boosting it in certain areas, combining it with other textures, or if Detail is set to 0, replacing the Perlin Noise map altogether.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Fast Noise to only those pixels within the mask.

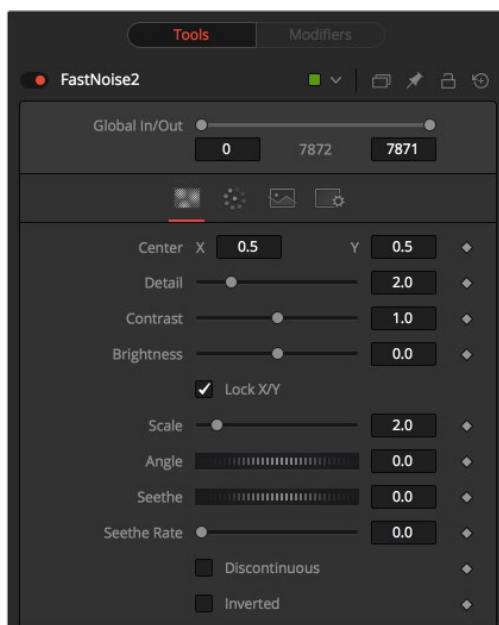
Basic Node Setup

The Fast Noise node is used to generate images for other nodes to take advantage of. For example, below the Fast Noise node is used as a bitmap source in the Particle Emitter.



A Fast Noise node used as a bitmap source for a Particle Emitter

Inspector



Fast Noise controls

Noise Tab

The Noise tab controls the shape and pattern of the noise for the Fast Noise node.

Discontinuous

Normally, the noise function interpolates between values to create a smooth continuous gradient of results. Enable this checkbox to create hard discontinuity lines along some of the noise contours. The result will be a dramatically different effect.

Inverted

Select this checkbox to invert the noise, creating a negative image of the original pattern. This is most effective when Discontinuous is also enabled.

Center

Use the Center coordinate control to pan and move the noise pattern.

Detail

Increase the value of this slider to produce a greater level of detail in the noise result. Larger values add more layers of increasingly detailed noise without affecting the overall pattern. High values take longer to render but can produce a more natural result.

Brightness

This control adjusts the overall brightness of the noise map, before any gradient color mapping is applied. In Gradient mode, this has a similar effect to the Offset control.

Contrast

This control increases or decreases the overall contrast of the noise map, prior to any gradient color mapping. It can exaggerate the effect of the noise and widen the range of colors applied in Gradient mode.

Lock and Scale X/Y

The size of the noise map can be adjusted using the Scale slider, changing it from gentle variations over the whole image to a tighter overall texture effect. The Scale slider can be separated into independent X- and Y-axis scale sliders by clicking on the Lock X/Y checkbox immediately above, which can be useful for a brushed-metal effect.

Angle

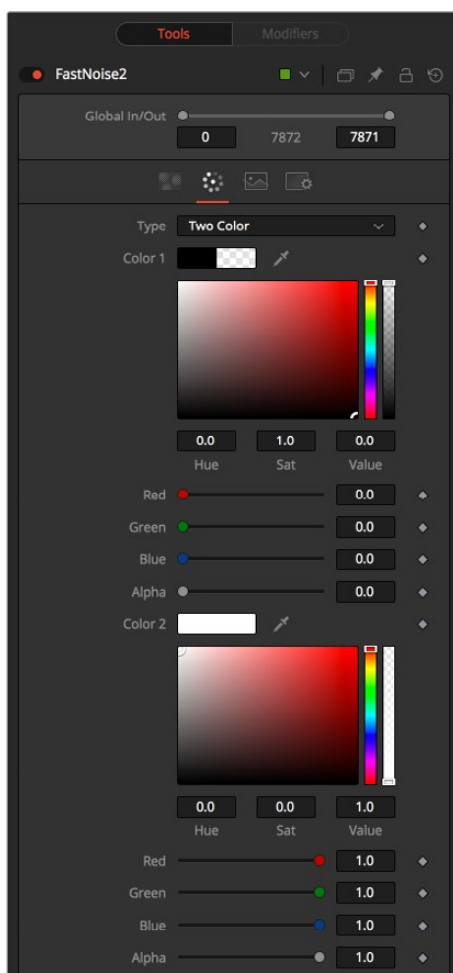
Use the Angle control to rotate the noise pattern.

Seethe

Adjust this thumbwheel control to interpolate the noise map against a different noise map. This will cause a crawling shift in the noise, as if it was drifting or flowing. This control must be animated to affect the gradient over time, or you can use the Seethe Rate control below.

Seethe Rate

As with the Seethe control above, the Seethe Rate also causes the noise map to evolve and change. The Seethe Rate defines the rate at which the noise changes each frame, causing an animated drift in the noise automatically, without the need for spline animation.



Fast Noise node Color tab

Color Tab

The Color tab allows you to adjust the gradient colors used in the generated noise pattern.

Two Color

A simple two-color gradient is used to color the noise map. The noise function will smoothly transition from the first color into the second.

Gradient

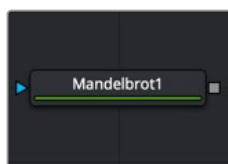
The Advanced Gradient control in Fusion is used to provide more control over the color gradient used with the noise map.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are duplicated in many Generator nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Mandelbrot [MAN]



The Mandelbrot node

Mandelbrot Node Introduction

This node creates an image pattern based on the Mandelbrot fractal theory set.

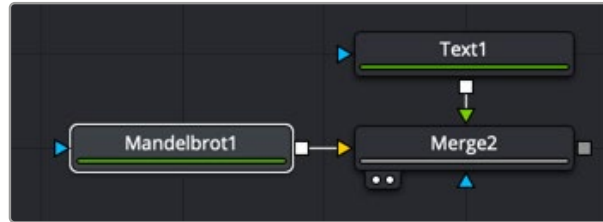
Inputs

The one input on the Mandelbrot node is for an effect mask to limit the area where the fractal noise is applied.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the fractals to only those pixels within the mask.

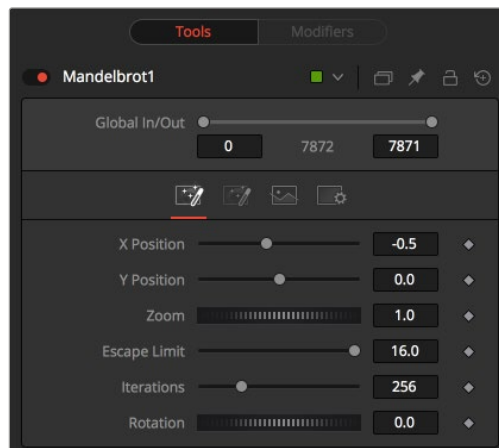
Basic Node Setup

The Mandelbrot node generates images that can be used for motion graphics and sci-fi effects. Below, it is used to generate motion graphics for the background of a title.



A Mandelbrot node applied as a background to the Merge node

Inspector



Mandelbrot node controls

Noise Tab

The Noise tab controls the shape and pattern of the noise for the Mandelbrot node.

Position X and Y

This chooses the image's horizontal and vertical position or seed point.

Zoom

Zoom magnifies the pattern in or out. Every magnification is recalculated so that there is no practical limit to the zoom.

Escape Limit

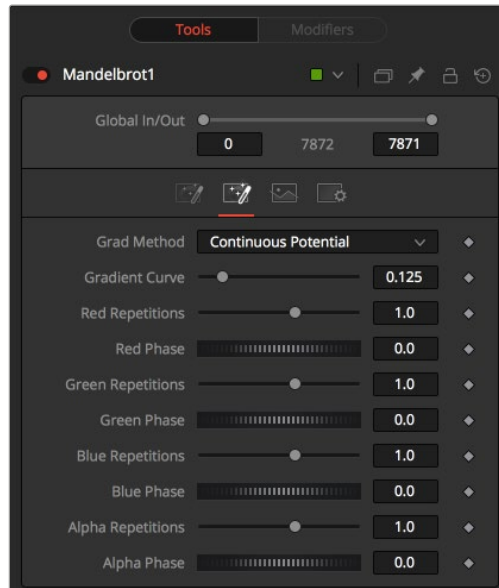
Defines a point where the calculation of the iteration is aborted. Low values lead to blurry halos.

Iterations

This determines the repetitiveness of the set. When animated, it simulates a growing of the set.

Rotation

This rotates the pattern. Every new angle requires recalculation of the image.



The Mandelbrot node Color tab

Color Tab

The Color tab allows you to adjust the gradient and repetition of the gradient colors for the generated pattern.

Grad Method

Use this control to determine the type of gradation applied at the borders of the pattern.

Continuous Potential

This causes the edges of the pattern to blend to the background color.

Iterations

This causes the edges of the pattern to be solid.

Gradient Curve

This affects the width of the gradation from the pattern to the background color.

R/G/B/A Phase and Repetitions

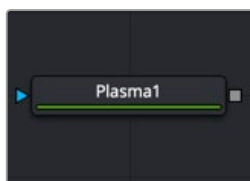
These controls set the color values of the pattern generators.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are duplicated in other generator nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Plasma [PLAS]



The Plasma node

Plasma Node Introduction

The Plasma node is a background generation node that uses four circular patterns to generate images similar to plasma. It is useful as a deforming pattern for the Shadow and Deform nodes and to create a variety of other useful shapes and patterns. It is similar to the Fast Noise node.

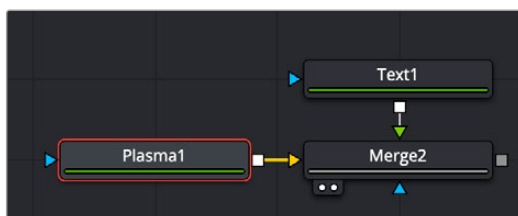
Inputs

The one input on the Plasma node is for an effect mask to limit the area where the plasma pattern is applied.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the plasma to only those pixels within the mask.

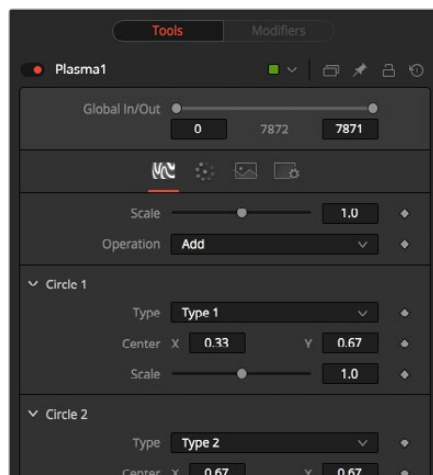
Basic Node Setup

Like the Mandelbrot node, the Plasma node generates images that can be used for motion graphics and sci-fi effects. Below, it is used to generate motion graphics for the background of a title.



A Plasma node applied as a background to a Merge node

Inspector



Plasma node Circles tab

Circles Tab

The Circles tab controls the shape and pattern generated by the Plasma node.

Scale

The Scale control is used to adjust the size of the pattern created.

Operation

The options in this menu determine the mathematical relationship among the four circles whenever they intersect.

Circle Type

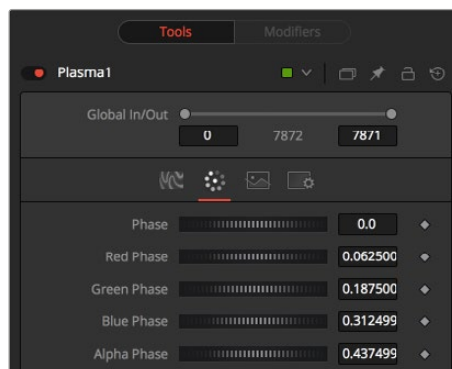
Select the type of circle to be used.

Circle Center

Report and change the position of the circle center.

Circle Scale

Determines the size of the circle to be used for the pattern.



Plasma node Color tab

Color Tab

The Color tab allows you to adjust the colors and location within the pattern of the colors for the generated plasma.

Phase

Phase changes the color phase of the entire image. When animated, this creates psychedelic color cycles.

R/G/B/A Phases

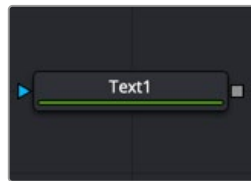
Changes the phase of the individual color channels and the Alpha. When animated, this creates color cycling effects.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are duplicated in many Generator nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Text+ [Ttxt+]



The Text+ node

Text+ Node Introduction

Fusion's Text+ node is an advanced character generator capable of multiple styles, 3D transformations, and several layers of shading. Text can be laid out to a user-defined frame, circle, or along a path.

Any TrueType, OpenType, or PostScript 1 font installed on the computer can be used to create text. Support for multibyte and Unicode characters allows text generation in any language, including right to left and vertically oriented text.

This node generates a 2D image. To produce extruded 3D text with optional beveling, see the Text 3D node.

Inputs

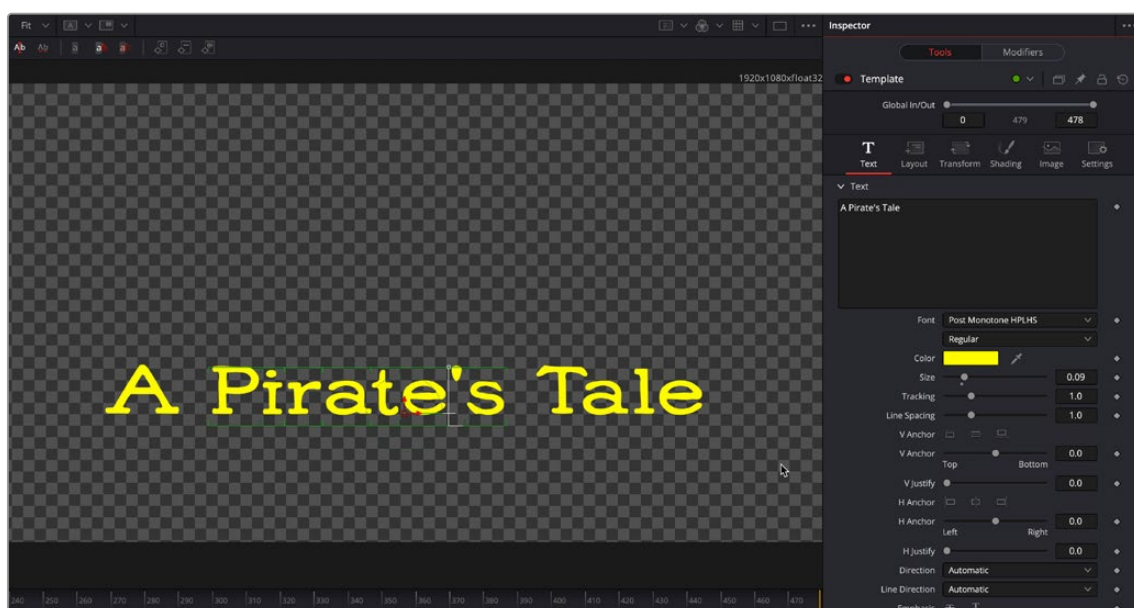
The one input on the Text+ node is for an effect mask to crop the text.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the text to only those pixels within the mask.

On-screen Controls

You can perform various editing actions directly on Text+ in the Viewer overlay, such as editing text, displaying a cursor, and selecting text.

To activate this feature, hover over the popup toolbar within the Fusion viewer and activate it, or alternatively, right-click on the View menu. This functionality is intuitive; when Text+ is selected, double-clicking anywhere in the Fusion viewer activates the in-viewer text editing mode.



The Text+ on screen viewer editing controls in the Fusion viewer.

Within the Fusion viewer, you'll also find pivot points for your Character, Word, and Line transforms. For easy access, a Pivot Points icon has been added to the popup toolbar, allowing you to toggle their visibility when adjusting rotation and scaling transformations.

The Z position-based sorting mode is a powerful addition to our toolkit. In contrast to the Distance mode, which sorts based on absolute distance from the camera, this mode sorts solely on the Z distance from the camera, irrespective of the relative orientation. With Z position, you can flexibly arrange overlapping text elements, ensuring precise layering and depth management.

Character-level styling has been refined to facilitate more intuitive offsets and translation transformations. This enhancement reduces the need for manual positioning and ensures a consistent application across multiple separate character selections.

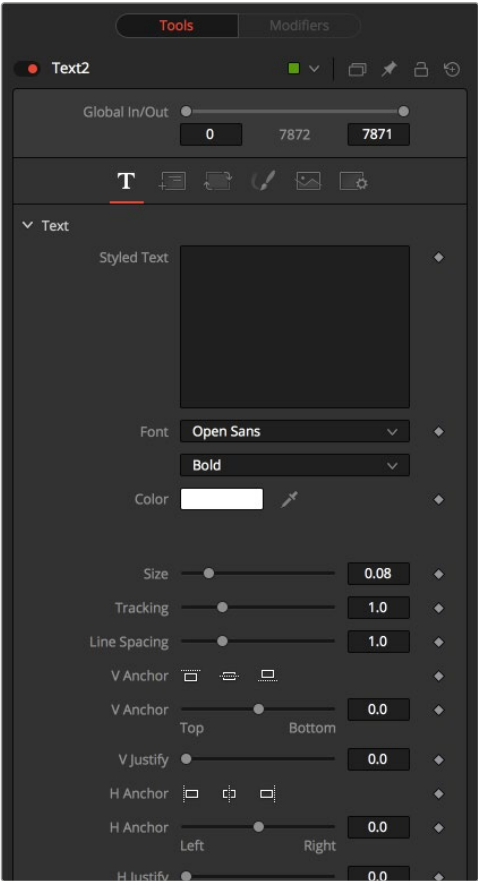
Basic Node Setup

The Text+ node can be combined with many other nodes to create motion graphics. Below it is connected as the foreground of a Merge node.



Text+ node connected as the foreground to a Merge node

Inspector



Text+ Text tab controls

Text Tab

The Text tab in the Inspector is divided into three sections: Text, Advanced Controls, and Tab Spacing. The Text section includes parameters that will be familiar to anyone who has used a word processor. It includes commonly used text formatting options. The Advanced controls are used for kerning options, and the Tab Spacing is used to define the location and alignment of tabs in the layout.

Styled Text

The edit box in this tab is where the text to be created is entered. Any common character can be typed into this box. The common OS clipboard shortcuts (Command-C or Ctrl-C to copy, Command-X or Ctrl-X to cut, Command-V or Ctrl-V to paste) will also work; however, right-clicking in the edit box displays a custom contextual menu. More information on these modifiers can be found at the end of this section.

The Styled Text contextual menu includes the following options:

- **Animate:** Used to animate the text over time.
- **Character Level Styling:** Used to change the font, color, size and transformations of individual characters or words through the Modifiers tab.
- **Comp Name:** Places the name of the composition in the Styled text box for creating slates.
- **Follower:** A text modifier used to ripple animation across each character of the text.
- **Publish:** Publishes the text for connection to other text nodes.
- **Text Scramble:** A text modifier used to randomize the characters in the text.
- **Text Timer:** A text modifier used to display a countdown or the current date and time.
- **Time Code:** A text modifier used to display Time Code for the current frame.
- **Connect To:** Used to connect the text to the published output of another node.

Font

Two Font menus are used to select the font family and typeface, such as Regular, Bold, and Italic.

Color

Sets the basic fill color of the text. This is the same control displayed in the Shading tab color swatch.

Size

This control is used to increase or decrease the size of the text. This is not like selecting a point size in a word processor. The size is relative to the width of the image.

Tracking

The Tracking parameter adjusts the uniform spacing between each character of text.

Line Spacing

Line Spacing adjusts the distance between each line of text. This is sometimes called leading in word-processing applications.

V Anchor

The vertical anchor controls consist of three buttons and a slider. The three buttons are used to align the text vertically to the top of the text, middle of the text, or bottom baseline. The slider can be used to customize the alignment. Setting the vertical anchor will affect how the text is rotated as well as the location for line spacing adjustments. This control is most often used when the Layout type is set to Frame in the Layout tab.

V Justify

The vertical justify slider allows you to customize the vertical alignment of the text from the V Anchor setting to full justification so it is aligned evenly along the top and bottom edges. This control is most often used when the Layout type is set to Frame in the Layout tab.

H Anchor

The horizontal anchor controls consist of three buttons and a slider. The three buttons justify the text alignment to the left edge, middle, or right edge of the text. The slider can be used to customize the justification. Setting the horizontal anchor will affect how the text is rotated as well as the location for tracking (leading) spacing adjustments. This control is most often used when the Layout type is set to Frame in the Layout tab.

H Justify

The horizontal justify slider allows you to customize the justification of the text from the H Anchor setting to full justification so it is aligned evenly along the left and right edges. This control is most often used when the Layout type is set to Frame in the Layout tab.

Direction

This menu provides options for determining the Direction in which the text is to be written, either horizontally or vertically in either direction. This allows certain Asian languages to flow properly during animation.

Line Direction

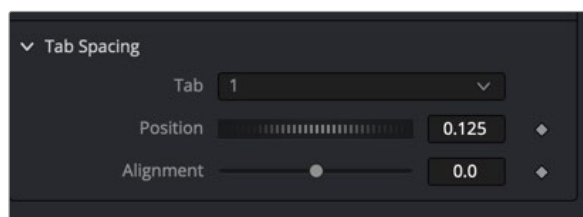
These menu options are used to determine the text flow from top to bottom, bottom to top, left to right, or right to left.

Underline and Strikeout

These buttons apply the addition of emphasis styles to the text.

Write On

This range control is used to quickly apply simple Write On and Write Off effects to the text. To create a Write On effect, animate the End portion of the control from 1 to 0 over the length of time required. To create a Write Off effect, animate the Start portion of the range control from 0 to 1.



Tab Spacing

Tab Spacing

The controls in the Tabs section are used to configure the horizontal screen positions of eight separate tab stops. Any tab characters in the text will conform to these positions.

You can add tabs directly in the Styled Text input as you type. You can also add tabs by copying from another document, such as Text on macOS or Notepad on Windows, and paste it into the text box.

Position

This control is used to set the horizontal position of the tab in the frame. The values range from -0.5 to 0.5, where 0 is the center. The position of the tab will be indicated in the viewer by a thin vertical white line when the Text node is selected. At the top of each tab line in the viewer is a handle. The handle can be used to position the tab manually.

Alignment

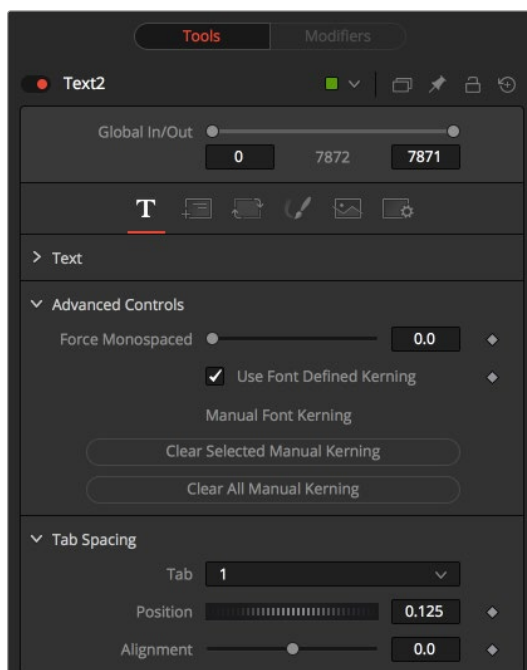
Each tab can be set either left aligned, right aligned, or centered. This slider ranges from -1.0 to 1.0, where -1.0 is a left-aligned tab, 0.0 is a centered tab and 1.0 is a right-aligned tab. Clicking the tab handles in the viewer will toggle the alignment of the tab among the three states.

Reading Direction

These options allow you to set the reading direction of the text, either automatically or manually. You can specify Left to Right languages like English, German, etc. or Right to Left languages like Arabic and Hebrew.

Force Monospaced

This slider control can be used to override the kerning (spacing between characters) defined in the font. Setting this slider to zero (the default value) will cause Fusion to rely entirely on the kerning defined with each character. A value of one will cause the spacing between characters to be completely even, or monospaced.



Advanced font controls

Use Font Defined Kerning

This enables kerning as specified in the TrueType font and is on by default.

Use Ligatures

If your font supports ligatures, you can activate them here by choosing All Scripts. Ligatures combine individual letters into single glyphs, like **ff** and **fl**. If you're animating individual text letters, often you want the ligature letters separated individually rather than as a single glyph, so None is the default for

Latin characters. Ligatures are required to render some languages like Arabic correctly, and use the Non-Latin setting.

Style Can Split Ligatures

Check this box If your font supports the option to split ligatures.

Stylistic Set

If your font includes stylistic sets, you can select them in the drop-down menu.

Font Features

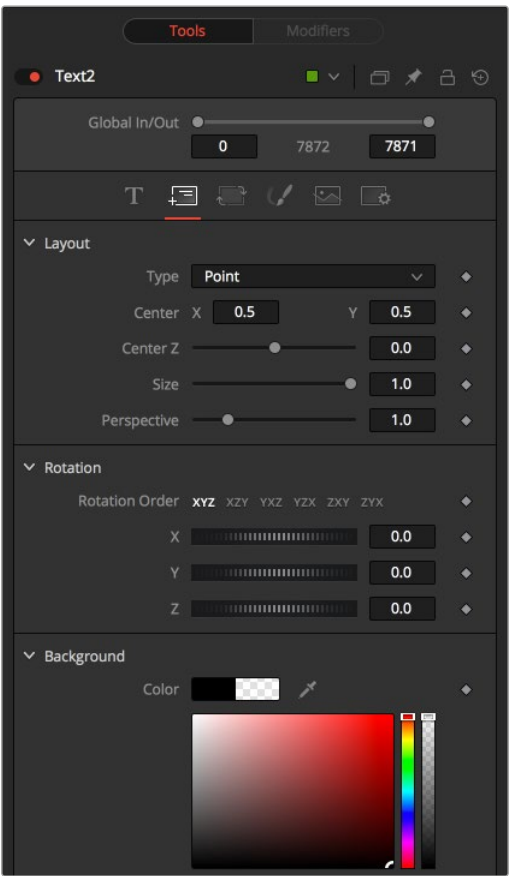
This allows you to set OpenType 4 letter tags to activate certain font features. For example, "smcp" will show **SMALL CAPITALS**, and "frac" will display fractions as ½ instead of 1/2. Not all features are supported by a particular font. A full list of OpenType feature codes can be found here: <https://docs.microsoft.com/en-us/typography/opentype/spec/featurelist>

Manual Font Kerning/Placement

Right-clicking on this label will display a contextual menu that can be used to animate the kerning of the text. See the “Toolbar” section of this node later in this chapter for details on manual kerning.

Layout Tab

The controls used to position the text are located in the Layout tab. One of four layout types can be selected using the Type drop-down menu.



Text+ Layout tab controls

- **Point:** Point layout is the simplest of the layout modes. Text is arranged around an adjustable center point.
- **Frame:** Frame layout allows you to define a rectangular frame used to align the text. The alignment controls are used for justifying the text vertically and horizontally within the boundaries of the frame.
- **Circle:** Circle layout places the text around the curve of a circle or oval. Control is offered over the diameter and width of the circular shape. When the layout is set to this mode, the Alignment controls determine whether the text is positioned along the inside or outside of the circle's edge, and how multiple lines of text are justified.
- **Path:** Path layout allows you to shape your text along the edges of a path. The path can be used simply to add style to the text, or it can be animated using the Position on Path control that appears when this mode is selected.

Center X, Y, and Z

These controls are used to position the center of the layout element in space. X and Y are onscreen controls, and Center Z is a slider in the node controls.

Size

This slider is used to control the scale of the layout element.

Perspective

This slider control is used to add or remove perspective from the rotations applied by the Angle X, Y, and Z controls.

Rotation

Rotation consists of a series of buttons allowing you to select the order in which 3D rotations are applied to the text. Angle dials can be used to adjust the angle of the Layout element along any axis.

Width and Height

The Width control is visible when the Layout mode is set to Circle or Frame. The Height control is visible only when the Layout mode is set to Frame. Width and Height controls are used to adjust the dimensions and aspect of the Layout element.

Fit Characters

This menu control is visible only when the Layout type is set to Circle. This menu is used to select how the characters are spaced to fit along the circumference.

Position on Path

The Position on Path control is used to control the position of the text along the path. Values less than 0 or greater than 1 will cause the text to move beyond the path in the same direction as the vector of the path between the last two keyframes.

Background Color

The text generated by this node is normally rendered with a transparent background. This Color Picker control can be used to set a background color.

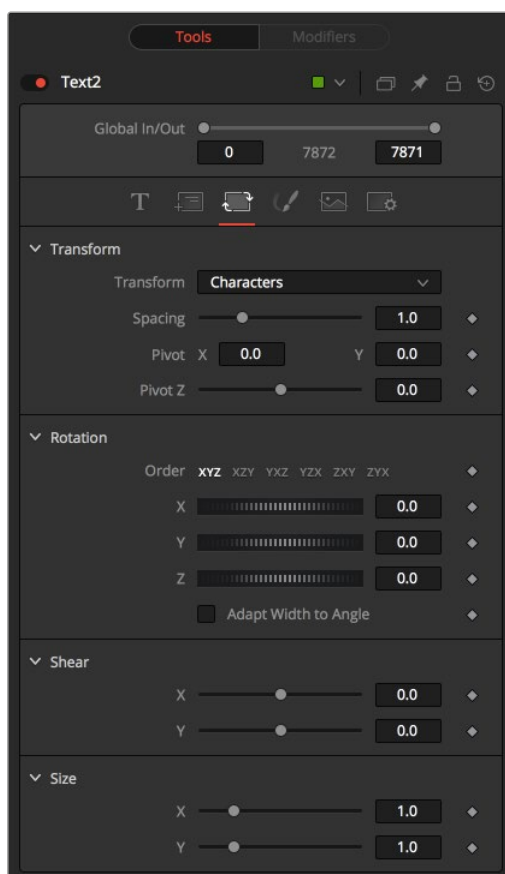
Right-Click Here for Shape Animation

This label appears only when the Layout type is set to Path. It is used to provide access to a contextual menu that provides options for connecting the path to other paths in the node tree, and animating the shape of the path over time.

For more information, see *Chapter 73, "Animating with Motion Paths,"* in the DaVinci Resolve Reference Manual, or Chapter 11 in the Fusion Reference Manual.

Transform Tab

The Transform tab is used to move, rotate, shear and scale text based on a character, word, or line.



Text+ Transform tab controls

Transform

The Transform menu is used to determine the portion of the text affected by the transformations applied in this tab. Transformations can be applied to line, word, and character levels simultaneously. This menu is only used to keep the visible controls to a reasonable number.

- **Characters:** Each character of text is transformed along its own center axis.
- **Words:** Each word is transformed separately on the word's center axis.
- **Lines:** Each line of the text is transformed separately on that line's center axis.

Spacing

The Spacing slider is used to adjust the space between each line, word, or character. Values less than 1 will usually cause the characters to begin overlapping.

Pivot X, Y, and Z

This provides control over the exact position of the axis. By default, the axis is positioned at the calculated center of the line, word, or character. The Axis control works as an offset, such that a value of 0.1, 0.1 in this control would cause the axis to be shifted downward and to the right for each of the text elements. Positive values in the Z-axis slider will move the axis of rotation further along the axis (away from the viewer). Negative values will bring the axis of rotation closer.

Rotation

These buttons are used to determine the order in which transforms are applied. X, Y, and Z would mean that the rotation is applied to X, then Y, and then Z.

X, Y, and Z

These controls can be used to adjust the angle of the text elements in any of the three dimensions.

Shear X and Y

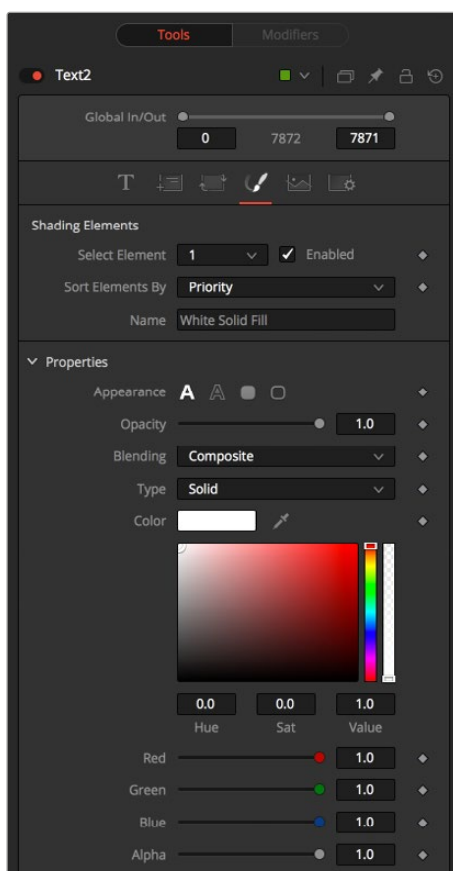
Adjust these sliders to modify the slanting of the text elements along the X- and Y-axis.

Size X and Y

Adjust these sliders to modify the size of the text elements along the X- and Y-axis.

Shading Tab

The Shading tab provides controls to adjust the shading, texture, and softness of the text. Transformations can be controlled from this tab as well, applying additional transformations to as many as eight separate text shading elements independently.



Text+ Shading tab controls

Shading Element

The eight number values in the menu are used to select the element affected by adjustments in this tab.

Enabled

Select this checkbox to enable or disable each layer of shading elements. Element 1, which is the fill color, is enabled by default. The controls for a shading element will not be displayed unless this checkbox is selected.

Sort By

This menu allows you to sort the shading elements by number priority, with 1 being the topmost element and 8 being the bottommost element, or Z depth, based on the Z Position parameter.

Name

This text label can be used to assign a more descriptive name to each shading element you create.

Appearance

The four Appearance buttons determine how the shading element is applied to the text. Different controls will appear below depending on the appearance type selected.

- **Text Fill:** The shading element is applied to the entire text. This is the default mode.
- **Text Outline:** The shading element is drawn as an outline around the edges of the text.
- **Border Fill:** The shading element fills a border surrounding the text. Five additional controls are provided with this shading mode.
- **Border Outline:** The Border Outline mode draws an outline around the border that surrounds the text. It offers several additional controls.

Opacity

The Opacity slider controls the overall transparency of the shading element. It is usually better to assign opacity to a shading element than to adjust the Alpha of the color applied to that element.

Blending

This menu is used to select how the renderer deals with an overlap between two characters in the text.

- **Composite:** Merges the shading over the top of itself.
- **Solid:** Sets the pixels in the overlap region to opaque.
- **Transparent:** Sets the pixels in the overlap region to transparent.

Thickness

(Outline only) Thickness adjusts the thickness of the outline. Higher values equal thicker outlines.

Adapt Thickness to Perspective

(Outline only) Selecting this checkbox will cause your outline to become thinner where the text is farther away from the camera, and thicker where it is closer. This will create a more realistic outline for text transformed in 3D but takes significantly longer to render.

Outside Only

(Outline only) Selecting this checkbox will cause the outline to be drawn only on the outside edge of the text. By default, the outline is centered on the edge and partially overlaps the text.

Join Style

(Outline only) These buttons provide options for how the corners of the outline are drawn. Options include Sharp, Rounded, and Beveled.

Line Style

(Outline only) This menu offers additional options for the style of the line. Besides the default solid line, a variety of dash and dot patterns are available.

Level

(Border Fill only) This is used to control the portion of the text border filled.

- **Text:** This draws a border around the entire text.
- **Line:** This draws a border around each line of text.
- **Word:** This draws a border around each word.
- **Character:** This draws a border around each character.

Extend Horizontal and Extend Vertical

(Border only) Use this slider to change the dimensions of each border.

Round

(Border Fill and Border Outline only) This slider is used to round off the edges of the border.

Color Types

Besides solid shading, it is also possible to use a gradient fill or map an external image onto the text. This menu is used to determine if the color of the shading element is derived from a user-selected color or gradient, or if it comes from an external image source. Different controls will be displayed below depending on the Color Type selected.

- **Solid:** When the Type menu is set to Solid mode, color selector controls are provided to select the color of the text.
- **Image:** The output of a node in the node tree will be used to texture the text. The node used is chosen using the Color Image control that is revealed when this option is selected.
- **Gradient:** When the Type menu is set to Gradient, additional controls are displayed where the gradient colors can direction can be customized.

Using the Gradient Color Bar

The gradient color bar is used to select the colors for the gradient. The default two color stops set the start and end colors. You can change the colors used in the gradient by selecting the color stop, and then using the Eyedropper or color swatch to set a new color.

You can add, move, copy and delete color using the gradient bar.

To modify one of the colors, select the triangle below the color on the bar.

To add a color stop to the gradient bar:

- 1 Click anywhere along the bottom of the gradient bar.
- 2 Use the Eyedropper or color swatch to select the color for the color stop.

To move a color stop on the gradient bar:

Drag a color stop left or right along the bar.

To copy a color stop:

Hold Command (macOS) or Ctrl (Windows) while you drag a color stop.

To delete a color stop:

Drag the color stop up past the gradient bar.

Image Source

(Image Mode only) The Image Source menu includes three options for acquiring the image used to fill the text.

- **Tool:** Displays a Color image text field where you can add a tool from the node tree as the fill for text.
- **Clip:** Provides a Browse button to select a media file from your hard drive as the fill for text.
- **Brush:** Displays a Color Brush menu where you can select one of Fusion's paint brush bitmaps as the fill for text.

Color Image/Color Brush

(Image Mode only) The Color Image text box and Color Brush menu are used to select the tool or paint brush that will fill the text. In the Color Image text box, enter the name of the node in the node tree that will provide the image. You can type the name in with the keyboard, drag the node from the node tree into the text box, or right-click and select Connect To from the contextual menu to select the image to be used. For the Color Brush, select the Brush from the menu.

Image Sampling

(Image Mode only) This menu is used to select the sampling type for shading rendering and transformations. The default of Pixel shading is sufficient for 90% of tasks. To reduce detectable aliasing in the text, set the sampling type to Area. This is slower but may produce better-quality results. A setting of None will render faster, but with no additional sampling applied so the quality will be lower.

Image Edges

(Image Mode only) This menu is used to choose how transformations applied to image shading elements are handled when they wrap off the text's edges.

Shading Mapping

(Image Mode only) This menu is used to select whether the entire image is stretched to fill the text or scaled to fit, maintaining the aspect ratio but cropping part of the image as needed.

Mapping Angle

(Image and Gradient Modes only) This control rotates the image or gradient on the Z-axis.

Mapping Size

(Image and Gradient Modes only) This control scales the image or gradient.

Mapping Aspect

(Image and Gradient Modes only) This control vertically stretches or shrinks the image or gradient.

Mapping Level

(Image and Gradient Modes only) The Mapping Level menu is used to select how the image is mapped to the text.

- **Full Image:** Applies the entire image to the text.
- **Text:** Applies the image to fit the entire set of text.
- **Line:** Applies the image per line of text.
- **Word:** Applies the image per each word of text.
- **Character:** Applies the image per individual character.

Softness X and Y

These sliders control the softness of the text outline used to create the shading element. Control is provided for the X- and Y-axis independently.

Apply Softness to Fill Color

Selecting this checkbox will cause blur (softness) to be applied to the shading element. The effect is best seen when applied to a shading element using an external image.

Softness Glow

This slider will apply a glow to the softened portion of the shading element.

Softness Blend

This slider controls the amount that the result of the softness control is blended back with the original. It can be used to tone down the result of the soften operation.

Priority Back/Front

Only enabled when the Sort By menu is set to Priority, this slider overrides the priority setting and determines the layer's order for the shading elements. Slide the control to the right to bring an element closer to the front. Move it to the left to tuck one shading element behind another.

Offset X, Y, and Z

These controls are used to apply offset from the text's global center (as set in the Layout tab) for the shading elements. A value of X0.0, Y0.1 in the coordinate controls would place the shading element centered with 10 percent of the image further down the screen along the Y-axis. Positive values in the Z-Offset slider control will push the center further away from the camera, while positive values will bring it closer to the camera.

Pivot X, Y, and Z

These controls are used to set the exact position of the axis for the currently selected shading element. By default, the axis is positioned at the calculated center of the line, word, or character. The axis control works as an offset, such that a value of 0.1, 0.1 in this control would cause the axis to be shifted downward and to the right for the shading element. Positive values in the Z-axis slider will move the axis of rotation further along the axis (away from the viewer). Negative values will bring the axis of rotation closer.

Rotation X, Y, and Z

These controls are used to adjust the angle of the currently selected shading element in any of the three dimensions.

Shear X and Y

Adjust these sliders to modify the slanting of the currently selected shading element along the X and Y axis.

Size X and Y

Adjust these sliders to modify the size of the currently selected shading element along the X and Y axis.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are duplicated in many Generator nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Text+ Toolbar

When the Text node is selected, a toolbar will appear in the viewer. Each button is described below from left to right.



Text+ toolbar

Allow Typing in Viewer

When selected, you can type and edit text directly in the viewer. Clicking on the text in the viewer will move the insert cursor within the text. Left and Right Arrow keys will move the insert cursor between characters. Using the Up and Down Arrows will move between text lines.

Allow Manual Kerning

The Manual Kerning button overrides the automatic kerning normally applied to text. Clicking the small red handle at the bottom of any character selects the character for kerning. You can also draw a selection rectangle around the characters you want to kern. Once any characters are selected, hold down the Option (macOS) or Alt key (Windows) while pressing the Left or Right Arrow key to make small adjustments to the kerning of the selected characters. Hold the Alt + Shift (Windows) or Option + Shift (macOS) keys down while pressing arrow keys to move the character in larger increments.

To animate the position of each character, right-click on the control label Manual Font Kerning in the Inspector’s Advanced Controls and select Animate from the contextual menu. A new key will be set on the animation spline each time a character is moved. All characters are animated with the same spline, as with polyline mask animation.

No Text Outline

When this button is selected, it disables the drawing of any outline around the edges of the text. The outline is not a part of the text; it is an onscreen control used to help identify the position of the text. This is a three-way toggle with the Text Outline Outside Frame Only, and Show Always Text Outline buttons.

Text Outline Outside Frame Only

This button draws an outline around the edges of text, which is outside the visible frame. This is useful for locating text that has moved offscreen and is no longer rendering a visible result. This is a three-way toggle with the No Text Outline, and Show Always Text Outline buttons.

Show Always Text Outline

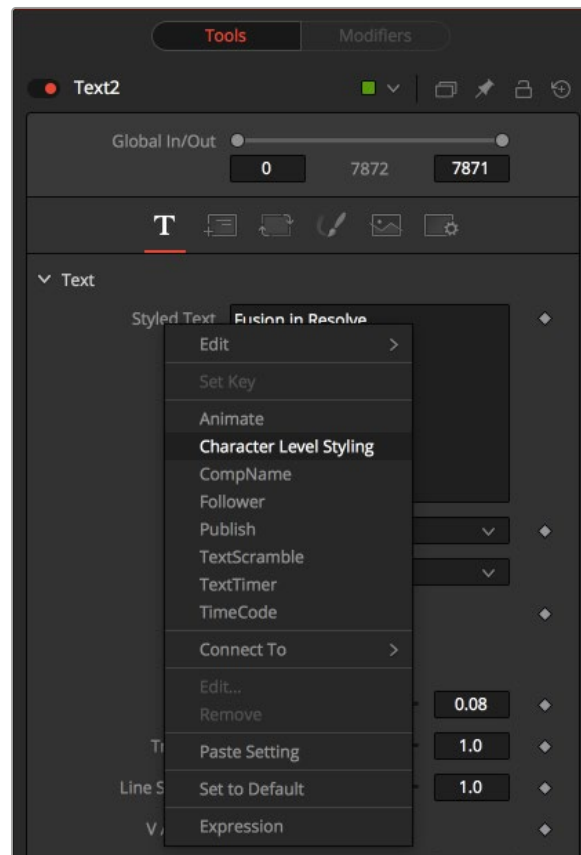
This button draws an outline around the edges of text, whether or not the text is visible within the frame. This is a three-way toggle with the No Text Outline, and Text Outline Outside Frame Only buttons.

Common Controls

Image and Settings Tabs

The Image and Settings tabs in the Inspector are duplicated in many Generator nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Text+ Modifiers



Text+ modifiers

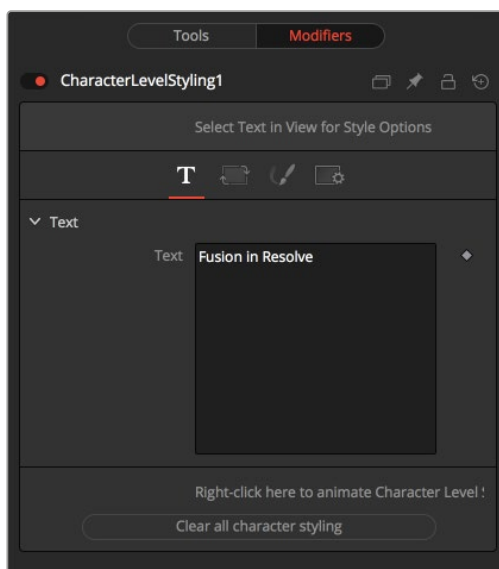
Text modifiers can be assigned by right-clicking in the Styled Text box and selecting a modifier from the contextual menu. Once a modifier is selected, its controls are found in the Modifiers tab at the top of the Inspector.

Character Level Styling

The Character Level Styling modifier works only on Text+ nodes. Once applied, individual characters can be selected directly in the viewer, and different text attributes can be applied to them using the controls in the Modifiers tab.

NOTE: Character Level Styling can only be directly applied to Text+ nodes, not to Text 3D nodes. However, styled text from a Text+ node can be applied to a Text 3D node by copying the Text+, right-clicking on the Text 3D, and choosing Paste Settings.

Inspector



Character Level Styling Modifiers tab

Text Tab

The Styled Text box in the Modifiers tab displays the same text in the Tools tab of the Text+ Inspector. However, individual characters you want to modify cannot be selected in the Styled Text box; they must be selected in the viewer. Once text is selected in the viewer, the Text tab includes familiar text formatting options that will apply only to the selected characters.

Clear Character Styling on Selection

All changes made to the currently selected characters will be reset.

Clear All Character Styling

All character attributes will be reset to their original values.

Transform Tab and Shading Tab

For details on these Text+ tabs, see the “Text+” section above.

Comp Name

The Comp Name sets the styled text to become the current Composition Name. This is quite useful to automate burn-ins for daily renderings. See also the TimeCode modifier. It can be applied by right-clicking in the Styled Text field of a Text+ node and selecting Comp Name.

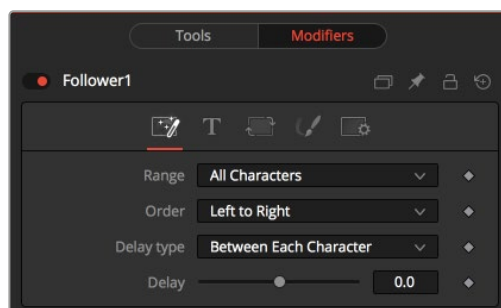
Controls

This modifier has no controls.

Follower

The Follower modifier allows sequencing text animations. The modifier is applied by right-clicking in the Styled Text field of a Text+ node and selecting Follower. In the Modifiers tab, you start by animating the parameters of the text (note that changing any parameter in the Modifiers tab will not be visible unless a keyframe is added.) Then, in the Timing tab you set the animation’s delay between characters.

Inspector



Follower Timing tab

Timing Tab

Once the text is animated using the controls in the Modifiers tab, the Timing tab is used to choose the direction and offset of the animation.

Range

The Range menu is used to select whether all characters should be influenced or only a selected range. To set the range, you can drag-select over the characters directly in the viewer.

Order

The Order menu determines in which direction the characters are influenced. Notice that empty spaces are counted as characters as well. Available options include:

- **Left to right:** The animation ripples from left to right through all characters.
- **Right to left:** The animation ripples from right to left through all characters.
- **Inside out:** The animation ripples symmetrically from the center point of the characters toward the margin.
- **Outside in:** The animation ripples symmetrically from the margin toward the center point of the characters.
- **Random but one by one:** The animation is applied to randomly selected characters but only influences one character at a time.
- **Completely random:** The animation is applied to randomly selected characters, influencing multiple characters at a time.
- **Manual curve:** The affected characters can be specified by sliders.

Delay Type

Determines what sort of delay is applied to the animation. Available options include:

- **Between Each Character:** The more characters there are in your text, the longer the animation will take to the end. A setting of 1 means the first character starts the animation, and the second character starts 1 frame later, the third character starts 1 frame after the second, and so on.
- **Between First and Last Character:** No matter how many characters are in your text, the animation will always be completed in the selected amount of time.

Clear All Character Styling

All character attributes will be reset to their original values.

Text Controls, Alignment, Transform, and Shading Tabs

In these tabs, the actual animation for the characters is done. Observe that simply changing a value in these tabs will have no influence at all. The value must be animated for the effect to show.

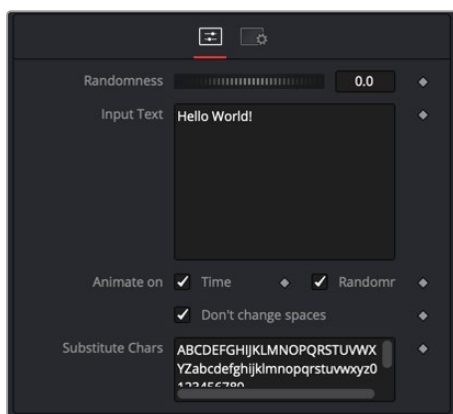
For a detailed description on the various parameters, see the Text+ node documentation.

Text Scramble

The Text Scramble randomly replaces the characters with others from a user definable set. It can be applied by right-clicking into the Styled Text field of a Text+ node and selecting Text Scramble.

The Controls for the Text Scramble are then adjusted in the Modifiers tab.

Inspector



Text Scramble modifier Controls tab

Controls Tab

The Controls tab in the Text Scramble modifier is used to enter text and scramble it using the Randomness control. The scrambled characters are taken from the Substitute Chars field at the bottom of the Inspector.

Randomness

Defines how many characters are exchanged randomly. A value of 0 will change no characters at all. A value of 1 will change all characters in the text. Animating this thumbwheel to go from 0 to 1 will gradually exchange all characters.

Input Text

This reflects the original text in the Text+ Styled Text field. Text can be entered either here or in the Text+ node.

Animate on Time

When enabled, the characters are scrambled randomly on every new frame. This switch has no effect when Randomness is set to 0.

Animate on Randomness

When enabled, the characters are scrambled randomly on every new frame, when the Randomness thumbwheel is animated.

This switch has no effect when Randomness is set to 0.

Don't Change Spaces

When enabled, spaces are not scrambled, allowing the length of the single words to stay the same, although their characters get scrambled around.

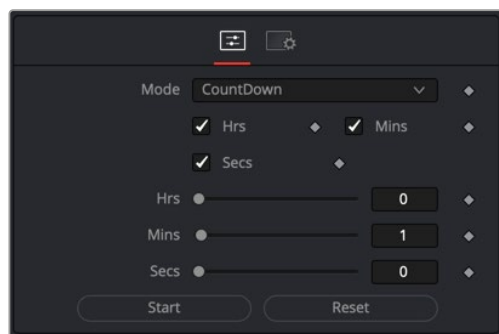
Substitute Chars

This field contains the characters used to scramble the text.

Text Timer

The Text Timer makes the Text+ node either a Countdown, a Timer, or a Clock. This is useful for onscreen real-time displays or to burn in the creation time of a frame into the picture. It can be applied by right-clicking in the Styled Text field of a Text+ node and selecting Text Timer.

Inspector



Text Timer modifier Controls tab

Controls Tab

The Controls tab for the Text Timer modifier is used to set up the type of time display that is generated by this modifier.

Mode

This menu sets the mode the timer is working in. The choices are Countdown, Timer, and Clock. In Clock mode, the current system time will be displayed.

Hrs, Mins, Secs (Checkboxes)

Defines which parts of the clock should be shown onscreen.

Hrs, Mins, Secs (Sliders)

Sets the start time for the Countdown and Timer mode.

Start

Starts the Counter or Timer. Toggles to Stop once the timer is running.

Reset

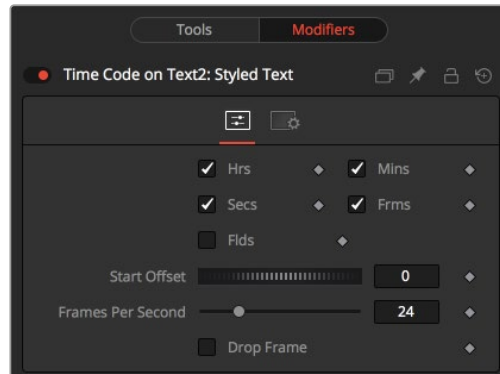
Resets the Counter and Timer to the values set by the sliders.

Time Code

The Time Code only works on Text+ nodes. It sets the Styled text to become a counter based on the current frame. This is quite useful for automating burn-ins for daily renderings.

It can be applied by right-clicking in the Styled Text field of a Text+ node and selecting Time Code.

Inspector



Time Code modifier Controls tab

Controls Tab

The Controls tab for the Time Code modifier is used to set up the time code display that is generated by this modifier.

Hrs, Mins, Secs, Frms, Flds

Activate or deactivate these options to customize the time code display to show hours, minutes, seconds, frames, and fields, respectively. Activating frames will only give you a plain frame counter.

Start Offset

Introduce a positive or negative offset to Fusion's current time to match up with existing time codes.

Frames per Second

This should match with your composition's FPS setting to provide accurate time measurement.

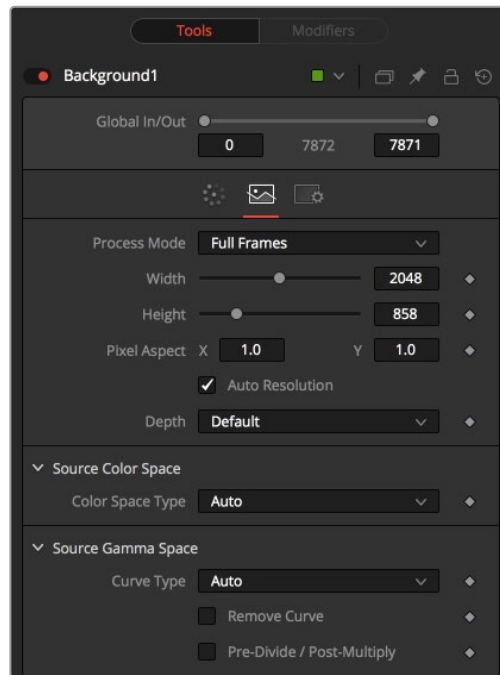
Drop Frame

Activate this checkbox to match the time code with footage that has drop frames—for example, certain NTSC formats.

The Common Controls

Nodes that generate images share a number of identical controls in the Inspector. This section describes controls that are common among Generator nodes.

Inspector



Background node Image tab

Image Tab

The controls in this tab are used to set the resolution, color depth, and pixel aspect of the image produced by the node.

Process Mode

Use this menu control to select the Fields Processing mode used by Fusion to render changes to the image. The default option is determined by the Has Fields checkbox control in the Frame Format preferences.

Global In and Out

Use this control to specify the position of this node within the project. Use Global In to specify which frame that starts the clip and Global Out to specify which frame ends the clip (inclusive) within the project's Global Range.

The node will not produce an image on frames outside this range.

Use Frame Format Settings

When this checkbox is selected, the width, height, and pixel aspect of the image created by the node will be locked to values defined in the composition's Frame Format preferences. If the Frame Format preferences change, the resolution of the image produced by the node will change to match. Disabling this option can be useful to build a composition at a different resolution than the eventual target resolution for the final render.

Width/Height

This pair of controls is used to set the Width and Height dimensions of the image to be created by the node.

Pixel Aspect

This control is used to specify the Pixel Aspect ratio of the created images. An aspect ratio of 1:1 would generate a square pixel with the same dimensions on either side (like a computer display monitor), and an aspect of 0.9:1 would create a slightly rectangular pixel (like an NTSC monitor).

NOTE: Right-click on the Width, Height, or Pixel Aspect controls to display a menu listing the file formats defined in the preferences Frame Format tab. Selecting any of the listed options will set the width, height, and pixel aspect to the values for that format, accordingly.

Depth

The Depth drop-down menu is used to set the pixel color depth of the image created by the Creator node. 32-bit pixels require 4X the memory of 8-bit pixels but have far greater color accuracy. Float pixels allow high dynamic range values outside the normal 0...1 range, for representing colors that are brighter than white or darker than black.

Source Color Space

You can use the Source Color Space menu to set the Color Space of the footage to help achieve a linear workflow. Unlike the Gamut tool, this doesn't perform any actual color space conversion, but rather adds the source space data into the metadata, if that metadata doesn't exist. The metadata can then be used downstream by a Gamut tool with the From Image option, or in a Saver, if explicit output spaces are defined there. There are two options to choose from:

- **Auto:** Automatically reads and passes on the metadata that may be in the image.
- **Space:** Displays a Color Space Type menu where you can choose the correct color space of the image.

Source Gamma Space

Using the Curve type menu, you can set the Gamma Space of the footage and choose to remove it by way of the Remove Curve checkbox when working in a linear workflow. There are three choices in the Curve type menu:

- **Auto:** Automatically reads and passes on the metadata that may be in the image.
- **Space:** Displays a Gamma Space Type menu where you can choose the correct gamma curve of the image.
- **Log:** Brings up the Log/Lin settings, similar to the Cineon tool.

For more information, see *Chapter 99, "Film Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 39 in the Fusion Reference Manual.

Remove Curve

Depending on the selected Gamma Space or on the Gamma Space found in Auto mode, the Gamma Curve is removed from, or a log-lin conversion is performed on, the material, effectively converting it to a linear output space.

Fast Noise Image Tab Options

The following controls are specific the Image tab in the Fast Noise Node.

Mask Map Inputs

These external connections allow you to use masks to control the value of the Noise Detail and Brightness controls individually for each pixel. This can allow some interesting and creative effects.

Noise Detail Map

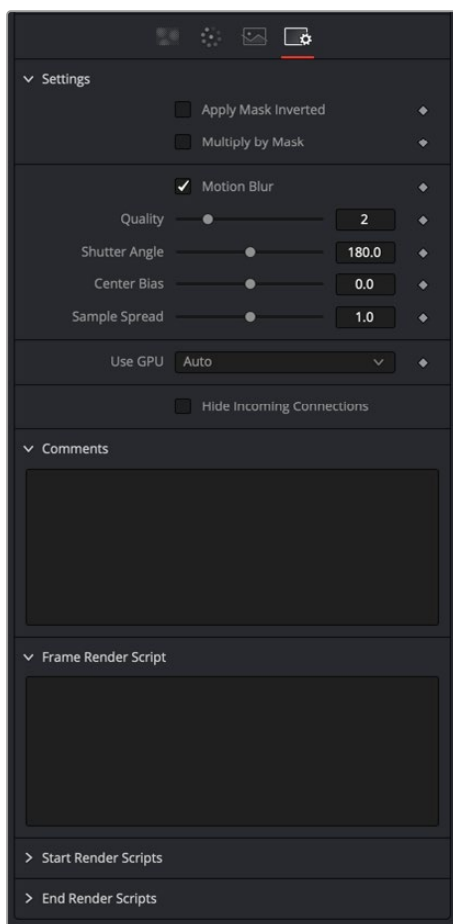
A soft-edged mask connected to the Noise Detail Map will give a flat noise map (zero detail) where the mask is black, and full detail where it is white, with intermediate values smoothly reducing in detail. It is applied before any gradient color mapping. This can be very helpful for applying maximum noise detail in a specific area, while smoothly falling off elsewhere.

Noise Brightness Map

A mask connected to this input can be used to control the noise map completely, such as boosting it in certain areas, combining it with other textures, or if Detail is set to 0, replacing the Perlin Noise map altogether.

Settings Tab

The Settings Tab in the Inspector can be found on every tool in the Color category. The Settings controls are even found on third-party Color-type plugin tools. The controls are consistent and work the same way for each tool, although some tools do include one or two individual options, which are also covered here.



Common Generator settings

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming. Normally, this will cause the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a Blur tool is deselected, the blur will first be applied to the image, and then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this will generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the mask channel's values. This will cause all pixels of the image not in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels will be used if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or *Chapter 18* in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID will be used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 will cause Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Use GPU

The user GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware, and Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Comments

The Comments field is used to add notes to a tool. Click in the field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.